

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460864.978 +/- 0.011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.048
Transit depth (Rp/Rs)^2	2.92 +/- 1.64 [%]
Orbital Inclination (inc)	89.21 +/- 2.36
Airmass coefficient 1 (a1)	0.922 +/- 0.039
Airmass coefficient 2 (a2)	0.068 +/- 0.055
Scatter in the residuals of the lightcurve fit is	4.23 %
Best Comparison Star	#1 - [396, 458]
Optimal Aperture	2.56
Optimal Annulus	10.12
Transit Duration (day)	0.052 +/- 0.013

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.171 $\pm$ 0.048 vs 0.1394 (deviation 0.49 σ)
Duration fit vs published	0.052 d vs 0.0983 d (deviation 2.66 σ)
Mid-time O-C	14.4 min
Depth SNR	2.7
Residual scatter	4.23 %

Light curve

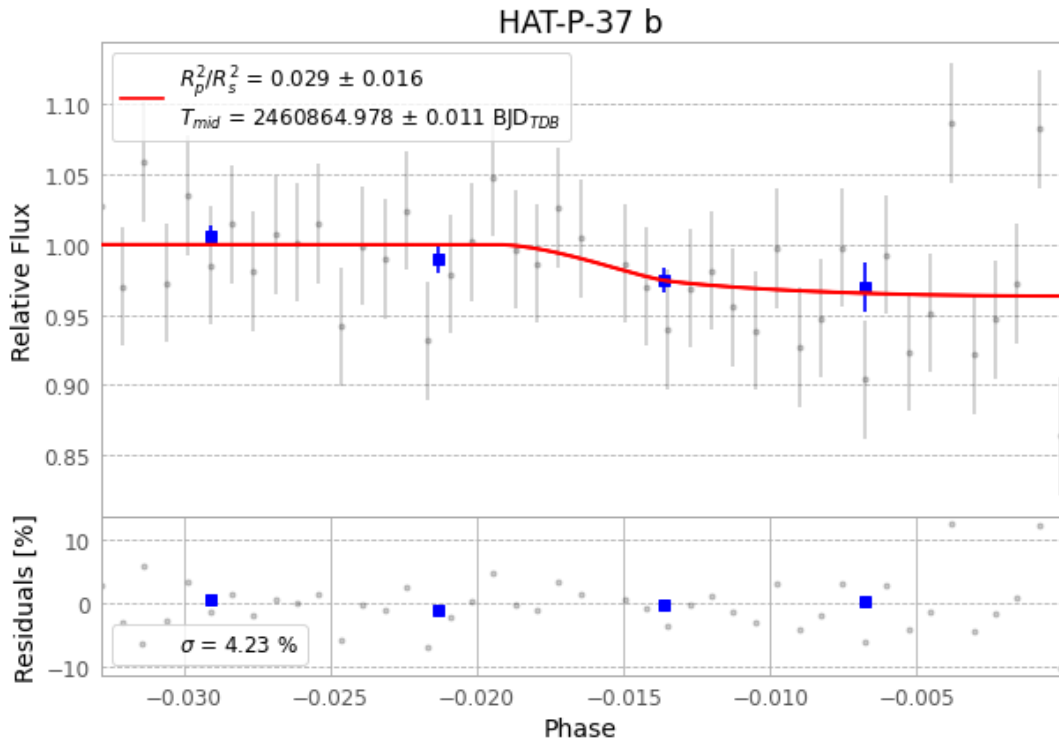


Figure 1 — FinalLightCurve_HAT-P-37 b_2025-07-08.png (SHA-256 82d7f5cf9aa25b20...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [302, 252], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 166d457c1797d43c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

45 FITS frames (30 MB), HATP-37250708091216.FITS → HATP-37250708112415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-250708.log (2df46a7a6e39...), FinalLightCurve_HAT-P-37 b_2025-07-08.png (82d7f5cf9aa2...), FinalLightCurve_HAT-P-37 b_2025-07-08.csv (b365ace1be4b...)

Compute resources

Wall time 112.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 23:21Z.